Chris Cameron

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I am a PhD researcher in electrophysiology (completion expected early 2027) and creator of Cubie. I also lecture part-time in undergraduate and postgraduate mechatronics, including computer architecture, C programming, circuit design, and embedded systems.

Cubie — GPU numerical integration

github.com/cubiepy/cubie

I built Cubie, an open-source Python ODE solver for consumer NVIDIA GPUs, to support electrophysiology simulations for a real-time application.

Draft, unpublished benchmarks show **10,000× speedup** over CPU implementations (SciPy, MATLAB) and **100×** over DiffEqGPU.jl.

- I used NVIDIA Nsight Compute to guide CUDA kernel optimisation, changing operation ordering, memory placement, and individual instructions.
- I tuned occupancy to fit cache sizes and adjusted loop unrolling to fit the instruction cache.
- I instrumented internal solver stages to verify numerical correctness throughout the integration.
- I review AI-generated implementations through pull requests and agent sessions, identifying errors and validating numerical behaviour in a specialised codebase.

Open-source contributions

numba-cuda / numba-cuda-mlir

My merged and open pull requests address compiler performance, typing and lowering correctness, AST transforms, and kernel performance controls.

SymPy and lcapy — minor edits.

Other projects

GPUODEBenchmarks github.com/ccam80/GPUODEBenchmarks

Performance benchmarks for GPU-based ODE solvers.

deep_zotero github.com/ccam80/deep-zotero

Local semantic search database for academic papers.